

Project Initialization and Planning Phase

Date	7 July 2026
Team ID	XXXXXXXXXX
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To develop an AI-powered Credit Card Approval Prediction system that predicts whether a credit card application will be approved using machine learning. The system aims to provide faster, more accurate, and transparent decisions while reducing manual effort..
Scope	The project analyzes applicant information such as income, employment status, credit score, existing debts, and financial history to predict credit card approval. It helps financial institutions improve efficiency and customer experience.
Problem Statement	
Description	The traditional credit card approval process is often slow, manual, and prone to inconsistencies. Applicants are frequently unaware of the reasons behind rejection, resulting in poor customer satisfaction

Impact	An intelligent prediction system reduces processing time, improves decision accuracy, minimizes financial risk, and increases customer satisfaction by providing faster and fairer approval decisions.
Proposed Solution	
Approach	Develop a machine learning model that analyzes applicant financial and personal information to predict the likelihood of credit card approval before manual verification.
Key Features	• AI-based credit card approval prediction • Real-time eligibility prediction • Fast and accurate decision making • Risk assessment using applicant data • Explainable predictions for transparency • Secure storage of applicant information • Easy-to-use web interface



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU	Intel Core i5/i7 (or Apple M1/M2 Mac), NVIDIA T4 GPU (optional for model training)
Memory	RAM	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD

Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Kaggle Credit Card Approval Dataset (~690 records), CSV format